

Project Proposal II – Cloud Computing

Cloud-Based Bug Tracking and Reporting System for Developers

Team number: 16

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1. Introduction

Software development teams rely on bug tracking systems to manage issues discovered during software testing and deployment. Existing systems such as issue trackers allow developers and users to report bugs, track their status, and collaborate on solutions.

The goal of this project is to develop a cloud-based bug tracking and reporting web application that allows users to submit, manage, and track software bugs in a centralized system. The platform will enable users to submit bug reports with details such as bug title, description, severity level, and screenshots. Developers and administrators will be able to monitor bug reports, assign them to responsible developers, and update the bug status accordingly.

The system will be deployed on a cloud platform using scalable infrastructure to ensure high availability and reliability. The application will demonstrate key cloud computing features including load balancing, auto scaling, secure access, and cloud storage. These features will allow the application to handle multiple user requests efficiently while maintaining system performance.

The scope of this project includes:

- Developing a web-based user interface for bug submission and tracking
- Implementing backend services for managing bug reports
- Deploying the application using cloud infrastructure
- Demonstrating scalability, reliability, and security using cloud services

The final system will allow users to interact with the bug tracking platform over the Internet while showcasing the use of modern cloud technologies. This aligns with the project objective of developing a functional cloud-based application that demonstrates scalability, reliability, elasticity, and security.

2. Proposed Design and Components

The proposed system will follow a cloud-based web application architecture consisting of a frontend interface, backend services, and cloud infrastructure components.

System Overview

Users will access the application through a web browser. Requests will be routed through a cloud load balancer which distributes traffic across multiple backend servers. The backend server processes the request, interacts with the database, and returns results to the user interface.

Screenshots uploaded during bug reporting will be stored in cloud object storage, while bug report information will be stored in a cloud database.

Key Components

1. Frontend Web Application
 - Provides user interface for bug submission and bug tracking
 - Allows users to create accounts and log in
 - Displays bug reports and status updates
2. Backend API Server
 - Handles business logic and communication with the database
 - Processes bug submissions and updates
 - Provides API endpoints for frontend communication
3. Load Balancer
 - Distributes incoming user traffic across multiple backend instances
 - Improves system availability and reliability
4. Auto Scaling Group
 - Automatically increases or decreases the number of server instances based on demand
 - Ensures the system can handle traffic spikes while optimizing resource usage
5. Cloud Database
 - Stores bug reports, user accounts, and bug status information
 - Enables persistent and reliable data storage
6. Cloud Storage
 - Stores uploaded screenshots related to bug reports
 - Allows users to attach visual evidence to bug submissions
7. Monitoring and Logging
 - Cloud monitoring services will track server performance and system health
 - Helps maintain reliability and detect issues quickly

Key Functional Features

The system will include the following features:

- Bug submission with title, description, severity level, and screenshot upload
- Bug status tracking (Open, In Progress, Resolved)
- Administrator dashboard for assigning bugs to developers
- Viewing bug history and updates
- Secure user authentication

Cloud Technologies

The application will utilize cloud services such as:

- Virtual machines for backend servers
- Elastic load balancing for traffic distribution
- Auto scaling services to dynamically adjust server capacity
- Cloud database services for persistent data storage
- Object storage services for file uploads
- Monitoring tools for system performance tracking

This architecture ensures the application is scalable, reliable, and elastic, which are important characteristics of cloud-based systems.

3. Schedule

Phase	Description	Start	End
Project Planning	Define project scope, system design, and team roles	Week 10	Week 11
Proposal Submission	Complete and submit project proposal	Week 11	Week 11
System Design	Design system architecture and database schema	Week 11	Week 12
Development	Implement frontend, backend, and cloud deployment	Week 12	Week 13
Testing	Test system functionality and cloud scalability	Week 13	Week 13
Documentation and Video	Prepare project report and demo video	Week 13	Week 13
Final Submission	Submit report, video demo, and source code	Week 13	Week 13

Resources required include cloud services such as virtual machines, storage services, database services, and monitoring tools.

4. Team members and roles

SN	Name of Team Member	Student ID	Responsible Components
1	Wong Jun Yu Kean	2301234	Frontend development (User interface, bug submission forms, dashboard)
2	Lee Wen Jie Brian	2301261	Backend API development and system logic
3	Lum Ko Sand	2301263	AI or automated bug classification and feature integration
4	Tan Si Han	2301264	Cloud infrastructure deployment, load balancing, and auto scaling
5	Wee Hong Ru Curtis	2301266	Database design, cloud storage integration, and security configuration